R80-R83 | Original release

One locked VAL checkpoint per run. Native/FP64 winner. ALL = equal weight across 18 tasks.

Model	TRAIN Top1 %	VAL Top1 %	TEST Top1 %	TEST cost	TEST regret %	VAL zero / winners
R77B	71.1767	70.8943	71.2252	11.345771	0.394014	30 / 2,648
R80A	71.2071	70.9728	71.2266	11.346230	0.397717	32 / 2,870
R80B	71.4382	70.9428	71.3151	11.345398	0.391480	25 / 1,297
R80C	71.6722	70.9571	71.1753	11.346086	0.397815	24 / 1,020
R80D	71.1681	70.8858	71.1767	11.345849	0.393588	29 / 2,410
R81B	71.1216	70.8558	71.1510	11.345798	0.393961	29 / 2,239
R82B	71.1452	70.9314	71.2780	11.345856	0.393312	26 / 2,114
R83	71.1309	70.8344	71.2067	11.345720	0.393365	27 / 2,159

R77B: Original R77 objective
R80A: CE only
R80B: Batch 512; matched sample exposures
R80C: Peak learning rate 3e-4
R80D: Four-view source embedding
R81B: Method-conditioned cross-instance AUC
R82B: Context-matched regret ranking
R83: Model-only source embedding

Raw costs cannot be compared across these two data releases. TRAIN-fitted SBS and Oracle are in metrics_by_problem.csv.

R84 | Matched NEW90k experiments

One locked VAL checkpoint per run. Native/FP64 winner. ALL = equal weight across 18 tasks.

Model	TRAIN Top1 %	VAL Top1 %	TEST Top1 %	TEST cost	TEST regret %	VAL zero / winners
R84A	71.2439	71.0852	71.2358	11.373843	0.394847	29 / 2,807
R84B	71.2597	71.0000	71.2235	11.373993	0.394592	25 / 1,471
R84C	71.3234	71.0235	71.2716	11.374011	0.397181	26 / 1,528
R84D	71.3191	70.9543	71.2173	11.373922	0.395155	23 / 1,067

R84A: NEW90k matched baseline
R84B: Direct attention selection
R84C: Original head plus focal penalty
R84D: Direct attention plus focal penalty

Raw costs cannot be compared across these two data releases. TRAIN-fitted SBS and Oracle are in metrics_by_problem.csv.

Zero selection already appears on TRAIN

Counts below come from actual saved predictions of the selected model, including R84D's matching update0043200 archive.

Model	TRAIN zero / winners	VAL zero / winners	TEST zero / winners	TRAIN-VAL Top1 pp
R77B	25 / 22,458	30 / 2,648	26 / 2,166	0.2824
R80A	24 / 16,374	32 / 2,870	24 / 2,059	0.2343
R80B	25 / 22,458	25 / 1,297	22 / 1,239	0.4954
R80C	21 / 16,013	24 / 1,020	25 / 1,330	0.7152
R80D	25 / 22,458	29 / 2,410	25 / 2,281	0.2823
R81B	24 / 22,293	29 / 2,239	26 / 2,327	0.2658
R82B	24 / 22,293	26 / 2,114	23 / 1,876	0.2138
R83	25 / 22,458	27 / 2,159	26 / 2,474	0.2965
R84A	26 / 25,627	29 / 2,807	25 / 2,246	0.1587
R84B	26 / 25,627	25 / 1,471	23 / 2,167	0.2597
R84C	23 / 18,259	26 / 1,528	22 / 1,405	0.2999
R84D	23 / 15,936	23 / 1,067	22 / 1,405	0.3648

A zero group means (problem, solver) with native winner_count > 0 and selected_count = 0.
Winner mass counts unique instances within that split. A method can leave the zero list after one incorrect pick.
These small TRAIN-VAL gaps do not prove the representation is sufficient or that the task has reached an upper bound.

Fixed target recovery | Correct winners, not selection quotas

Targets frozen from each release baseline's locked VAL: selected share <=0.1%, native winners >0.

Model	VAL correct / winners	Recall %	Rescued / lost	New false picks	False-pick regret pp
R77B	8 / 4358	0.184	0 / 0	0	0.000000
R80A	9 / 4358	0.207	5 / 4	3	0.000219
R80B	33 / 4358	0.757	25 / 0	23	0.003320
R80C	49 / 4358	1.124	42 / 1	43	0.006304
R80D	10 / 4358	0.229	4 / 2	2	0.000196
R81B	15 / 4358	0.344	8 / 1	8	0.001126
R82B	14 / 4358	0.321	7 / 1	9	0.001504
R83	11 / 4358	0.252	3 / 0	5	0.000381
R84A	11 / 4625	0.238	0 / 0	0	0.000000
R84B	32 / 4625	0.692	21 / 0	22	0.001277
R84C	21 / 4625	0.454	10 / 0	17	0.001660
R84D	56 / 4625	1.211	46 / 1	56	0.004474

Old and NEW90k target sets differ and must not be directly pooled.
Rescue = old missed winner now correctly selected. New false pick = newly selected target on a nonwinner.
No variant passes by merely shrinking the number of zero-selected methods.

R84 matched changes versus A

VAL determines acceptance; TEST is the final report for the already locked version.

Split	Model	Top1 delta pp	Corrected / harmed	Both wrong cheaper / c	Cost delta	Regret delta pp
VAL	R84B	-0.08519	1166 / 1235	561 / 548	0.000085979	0.001505
VAL	R84C	-0.06173	930 / 980	506 / 474	0.000390376	0.004802
VAL	R84D	-0.13086	1459 / 1565	751 / 722	0.000452389	0.004603
TEST	R84B	-0.01235	1211 / 1221	584 / 543	0.000150061	-0.000255
TEST	R84C	0.03580	1017 / 988	527 / 465	0.000168267	0.002335
TEST	R84D	-0.01852	1561 / 1576	779 / 737	0.000078974	0.000308

R84 all three variants fail the joint VAL gate: Top1 higher, mean_cost not higher, regret not higher.
R84C's +0.03580pp TEST Top1 comes with higher TEST cost and regret.
Do not promote a model or adjust its checkpoint/rule using TEST.

TRAIN winner discrimination | Selected checkpoints

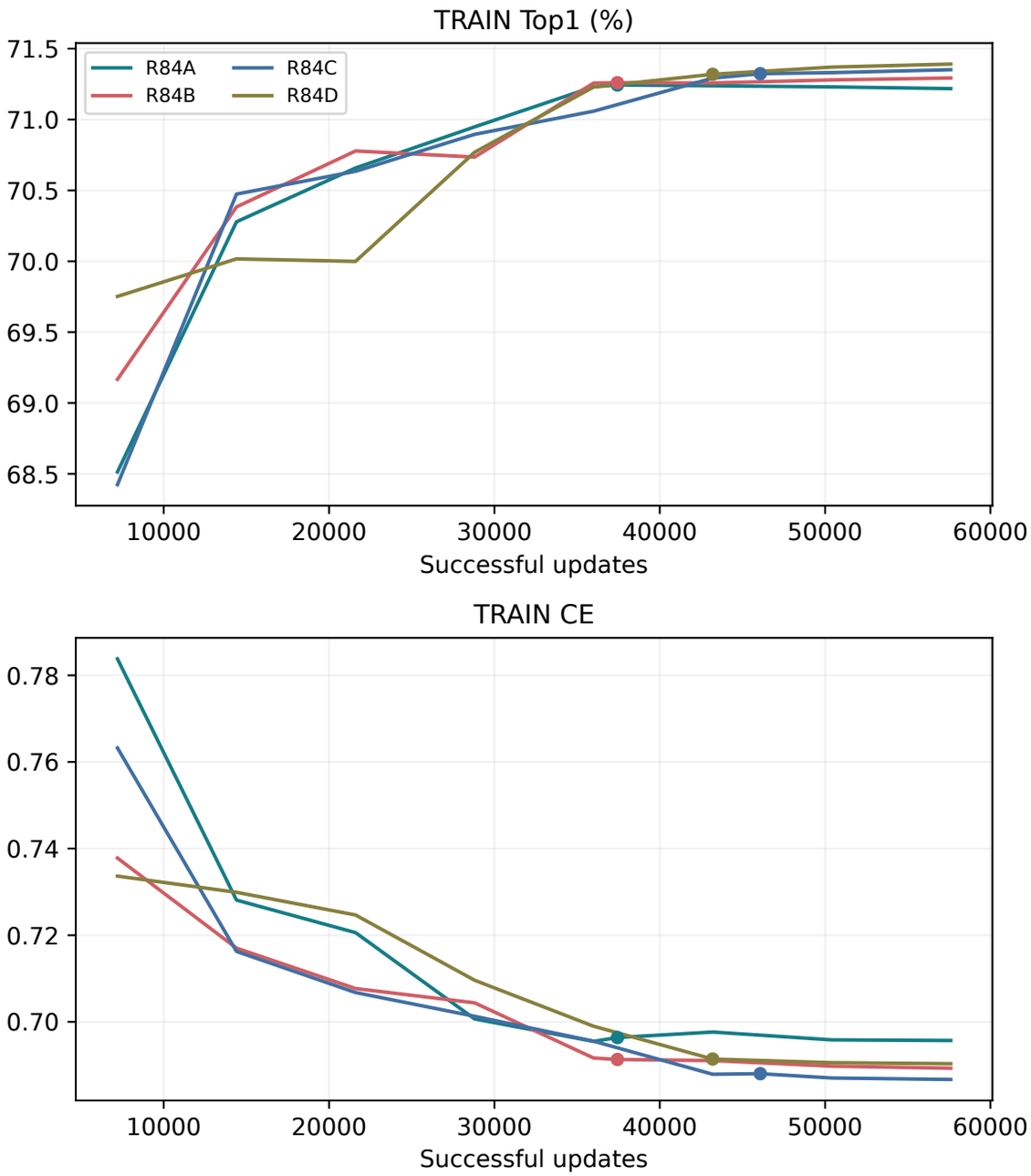
Large positive support still produces almost no correct argmax choices; this is observable before testing generalization.

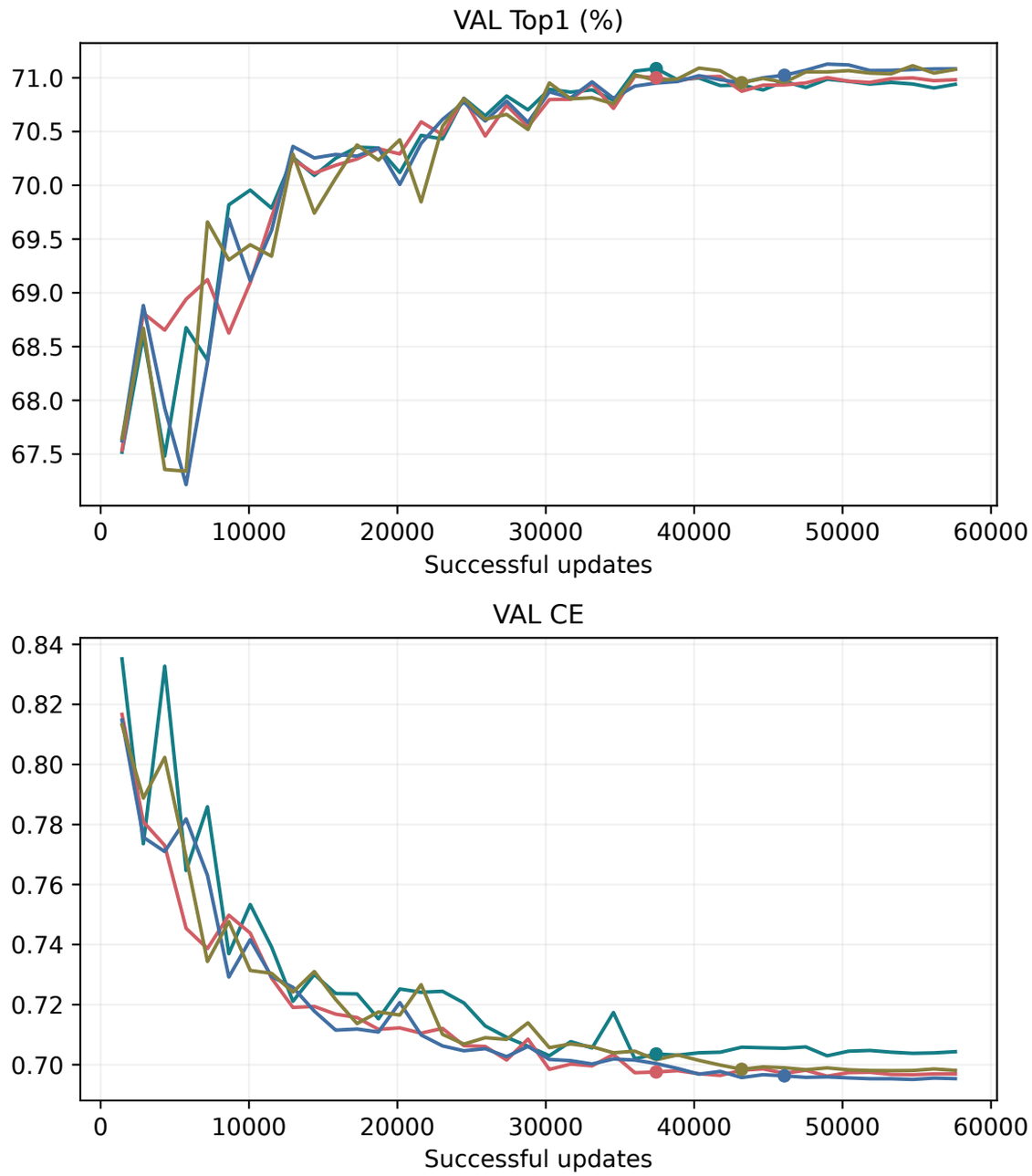
Model	Split	Method	Winners	Chosen	Correct	Recall %	Winner rank
R84A	TRAIN	TSP / DIFUSCO	6806	0	0	0.0000	3.642
R84A	TRAIN	TSP / LEHD	14279	15	7	0.0490	2.781
R84A	TRAIN	ATSP / MATPOENET	6476	0	0	0.0000	2.261
R84A	VAL	TSP / DIFUSCO	373	0	0	0.0000	3.611
R84A	VAL	TSP / LEHD	792	0	0	0.0000	2.677
R84A	VAL	ATSP / MATPOENET	373	0	0	0.0000	2.300
R84C	TRAIN	TSP / DIFUSCO	6806	10	4	0.0588	3.608
R84C	TRAIN	TSP / LEHD	14279	82	34	0.2381	2.721
R84C	TRAIN	ATSP / MATPOENET	6476	0	0	0.0000	2.300
R84C	VAL	TSP / DIFUSCO	373	0	0	0.0000	3.571
R84C	VAL	TSP / LEHD	792	5	4	0.5051	2.631
R84C	VAL	ATSP / MATPOENET	373	0	0	0.0000	2.327

R84A LEHD: 14,279 TRAIN winners but only 7 correct choices. ATSP MATPOENET: 6,476 winners, zero choices.
This supports investigating TRAIN discrimination / decision competition. It does not prove an accuracy ceiling.
Zero-list winner mass is not the same as all errors: one incorrect choice can remove a whole group from that list.

R84 optimization trajectories

All saved history shown for diagnosis; markers identify each model's previously locked checkpoint.





Locked TRAIN/VAL are close; final epochs remain around 71%. R84 does not reproduce severe training memorization seen in older binary-only experiments.

Evidence and implications

No new training / GPU inference / checkpoint selection. Single seed, same saved decision rules.

1. R80-R83 belong to OLD77904 (70,114/3,895/3,895 per task); R84 belongs to NEW90k (81,000/4,500/4,500). Compare within each release. Raw costs across releases are not a controlled gain.

2. The current locked R84 models have TRAIN Top1 around 71.2-71.3%, close to VAL. Zero selection is already present in TRAIN. These models do not show the near-100% TRAIN vs 50% holdout behavior of earlier binary readouts.

3. R84A/B TRAIN have 26 zero-with-winner groups / 25,627 winners. Focal C/D leave 23 groups / 18,259 and 15,936 winners. Reducing zero groups does not establish correct recovery or system improvement.

4. All three R84 variants have lower locked VAL Top1 and higher mean_cost/actual regret than A. Focal alone has slightly higher TEST Top1 but also higher TEST cost/regret; TEST does not override the failed VAL acceptance gate.

5. R81 global AUC, R82 matched ranking, R83 source filtering and the R80 optimizer/loss changes do not provide a clear system-level breakthrough. Same-method ranking evidence must not be confused with becoming the correct cross-method argmax.

6. Next action: prioritize method-specific TRAIN winner discrimination and the gap between ranking/probability support and final argmax. A generic longer-training/overfitting narrative is not justified for these locked R84 models. Any new proposal must improve VAL Top1 and reduce missed winner mass without increasing cost.

7. All runs use seed 2 only; no multi-seed significance claim. Native/FP64 winner ties follow the saved label. False picks can have zero extra cost if tied. All18 metrics are equal task means; mixed raw mean_cost is descriptive.

8. TEST has informed earlier research. This retrospective analysis does not select new checkpoints, thresholds, temperatures or per-task model mixtures. No GPU inference or new training was used.